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B.Tech. Degree IV Semester Special Supplementary Examination September 2014

CS 1403 COMPUTER ARCHITECTURE AND ORGANIZATION (2012 Scheme)

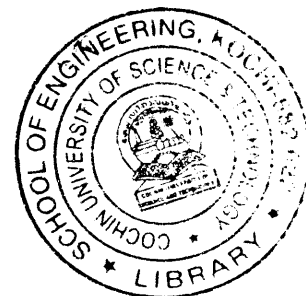
Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 x 5 = 40)

- I. (a) Explain basic operational concepts of a computer.
- (b) What is an assembly language? Write an assembly language program to find the sum of 'n' numbers.
- (c) Write the control sequence for executing a complete instruction Add (R3), R1.
- (d) Comment on booth algorithm with example.
- (e) Explain how memory interleaving speeds up the accessing of memory.
- (f) Distinguish between static and dynamic memory.
- (g) Explain the process of bus arbitration in DMA
- (h) Write short notes on :
 - (i) Synchronous bus.
 - (ii) Asynchronous bus



PART B

(4 x 15 = 60)

- II. Define addressing modes. Describe different addressing modes with examples. (15)
- OR**
- III. (a) Draw and explain functional units of a computer. (7)
- (b) Write short notes on stacks and subroutines. (8)
- IV. Describe and design the organization of hardwired control and micro programmed control. (15)
- OR**
- V. Explain the process of integer division with block diagram, algorithm and example. (15)
- VI. (a) What are cache memories? Explain various mapping functions. (12)
- (b) Briefly explain LRU page replacement algorithm. (3)
- OR**
- VII. What is meant by virtual memory? Explain virtual memory address translation. (15)
- VIII. Write short notes on:
 - (i) Vectored interrupts (5)
 - (ii) Interrupt nesting (5)
 - (iii) Daisy chaining. (5)
- OR**
- IX. Explain the different interfacing circuits. (15)